

Pareto Data Platform

What is it?

The Pareto Data Platform is an architecture blueprint for a data platform that fits 80% of the needs of medium size company with just a 20% of complexity

Assumptions

- The main use case for the data platform is Business Intelligence
- The company already use Microsoft Services, especially Microsoft Entra ID
- The major data sources run on-premise

Components

The Pareto Data Platform is composed of:

- Azure Data Factory as ETL Tool
- Azure Data Lake as Persistent Staging Area
- Azure Synapse as Data Warehouse
- Microsoft Power BI as BI Tool

Azure Data Factory as ETL Tool

Azure Data Factory is a cloud-based data integration service that enables you to automate data movement and transformation

Combined with Microsoft Integration Runtime, we will have an enterprise-grade ETL tool capable of extracting data from applications and databases running on-premises

Azure Data Lake as Persistent Staging Area

Azure Data Lake extends the capabilities of Azure Blob Storage to support hierarchical role-based access control while it remains extremely cost-effective

Using such as simple API and plain file-based storage will help us stay vendor-neutral and be able to migrate to any other cloud provider in the future

Storing the data in the Azure Data Lake before moving it to the data warehouse will create an additional backup layer